

SFML Project

prof. dr. Pieter Libin, Alvaro Vargas

March 23, 2026

1 Introduction

For the course *Statistical Foundations of Machine Learning* students will be marked on a project submitted no later than Sunday, 17 May 2025.

What is expected is an interactive research report in the form of a *Python Jupyter Notebook* addressing the assignment. It should be a **research** report, so you should state **three research questions** that you answer in the paper. It should be very clear what are the research questions, why they are important, and how you answer these questions. It should be an **interactive** report, so that the reader can easily reproduce your experiments. Python Jupyter notebooks provide a way to combine text, figures and Python code to run and report the experiments. We expect that you perform at least some experiments on 2- or 3-dimensional data points such that you can visualize the dataset and the decision boundary or regression function in your report. For more information on this, the blog "Easily visualize Scikit-learn models decision boundaries" shows how such visualisations can be done.

The notebook has to be self-contained¹, and the **reader should be able to understand the report without consulting other sources**, and to **run the notebook as it is**. This means that the datasets have to be included or are imported directly from the website where the dataset originates from.

2 Submission Modalities

You submit your project on Canvas. The **strict deadline** is Sunday, 17 May 2025 by midnight. Make sure to add the names of your group members and the master program in which they are enrolled, to your Python notebook.

3 Algorithms and datasets

You will submit **one Jupyter notebook** addressing three research questions. Make sure to mention the title of the project, the group's names and the master

¹The blog Learning Seattle's Work Habits from Bicycle Counts gives an example of a self-contained notebook.

program(s) in which the group members are enrolled. To address your research questions, you use two datasets, a synthetic and a real-world one, to be used for training, validation and testing purposes. For both classification and regression you use 2 different ML-algorithms.

You will work together in groups of three individuals, **please report your group composition on the following Google Sheet**([Google Sheet link](#)) before Tuesday, 31 March 2025. If you can't find a team mate, contact dr. Xavier Chesterman via e-mail to find a match (xavier.chesterman@vub.be).

Algorithms

You can use any ML-algorithm. It is not necessary that the algorithm was discussed during the classes or covered in the course material. Also: you do not have to implement the algorithms yourself. You can use a library like *scikit-learn*. In any case, it is important that you give a brief explanation of the ML-algorithm and the most important (hyper)parameters. In the Appendix, the explanation of the perceptron learning algorithm is given, to provide you with an example.

Datasets

1. *synthetic* dataset: You can use the python module *scipy.stats* for that purpose. You need to ensure that you have control over the data generation process, in order to set up meaningful experiments. The easiest way to do this, it's to generate the data yourself.
2. *real-world* dataset: You can find them at [Kaggle Datasets](#) the UCI ML repository at [UCI](#), etc. [Top Sources for ML Datasets](#) gives more information.

4 Research question examples

You are asked to apply ML-algorithms for classification and regression. Here, we give some examples of straightforward research questions:

1. How do the ML-algorithms compare in terms of in-sample error and out-of-sample error on the datasets used.
2. What is the impact of mislabeled training examples/outliers² on the performance?
3. What is the impact of class imbalances on the performance? In many cases we have about the same number of examples of each class. In some applications this is not the case, e.g. credit card fraud detection, intrusion detection, etc.

²A training example is mislabeled when it is put in the wrong class when labeling the training set.

4. What happens when classes overlap? How does that affect the performance? Which transformations to the feature space work best?
5. In the case of SVM, what is the impact of class overlap, class unbalance and mislabeled data on the number and location of support vectors?

Note that synthetic datasets are most useful to investigate the latter questions. You can control the percentage of mislabeled data, the class unbalance and the class overlap.

In the experiments, you are expected to use both synthetic and real-world datasets to answer at least one of the three research questions. To answer the other two research questions you can choose to use just one of the two datasets.

5 Use of Generative AI

The use of Generative AI (genAI) in this course is strictly limited to enhancing the clarity and readability of your written work. It may be used for grammar correction, sentence restructuring, and improving the flow of text. However, genAI cannot be used to generate ideas, conceptualize project components, or write any part of the required code. All project work must reflect your own original thought process and problem-solving skills. Any unauthorized use of genAI beyond these limitations will be considered a violation of academic integrity.

6 Evaluation

Your project will be evaluated based on the Python notebook you submit and the 10 minute presentation you give at a time slot during the exam period. Your work will be evaluated based on the originality and motivation of the research questions, the rigour with which you set up your experiments, the quality of your report and visualisations, and the description of the machine learning algorithms you used. You will also be evaluated on your presentation, taking into account the clarity, correctness of your presentation, and your ability to answer our questions after the presentation. To clarify our expectations with respect to the results, we constructed a rubric matrix that you can find on Canvas.

Appendix

A PLA - example description

We explain the perceptron learning algorithm (PLA).

Let

$$\{(\mathbf{x}_n, y_n) : \mathbf{x}_n \in \mathbb{R}^{d+1} \text{ and } y_n \in \{-1, 1\}\}$$

be the training set, $n = 1, \dots, N$. The *Perceptron Learning Algorithm* (PLA) is a classification algorithm that searches the hypothesis space

$$\mathcal{H}_{PLA} = \{h \in \mathcal{H} : h(x) = \text{sign}(\mathbf{w}^T \mathbf{x})\}$$

for a hypothesis $g(\mathbf{x})$ that classifies all $\mathbf{x}_n$ correctly. Note that we use the dummy feature $x_0 = 1$. The real features are $x_1, \dots, x_d$. As a result the hyperplane $h(\mathbf{w})$ goes through the origin $\mathbf{0}$ and is completely characterized by the weight vector $\mathbf{w}$ orthogonal to that hyperplane.

The final hypothesis $g(\mathbf{x})$ is obtained by iteratively updating $\mathbf{w}$ until all training examples are classified correctly. When the training set is *linearly separable* this is always possible.

PLA proceeds as follows. First, initialize the weight vector: $\mathbf{w} \leftarrow \mathbf{w}_0$. Next, select a misclassified point $\mathbf{x}_k$ and update the weight vector

$$\mathbf{w} \leftarrow \mathbf{w} + \eta y_k \mathbf{x}_k$$

until all examples are classified correctly, the parameter $\eta > 0$ is the learning rate. The number of iterations needed depends on the learning rate $\eta > 0$. The parameter η can be tuned using validation.

A variant called the pocket algorithm is used when the training set is not linearly separable. The number of iterations N is determined beforehand and the best hypothesis evaluated during the run is returned. The best hypothesis minimizes the number of misclassifications.